



PRESIDENCY COLLEGE

(AUTONOMOUS)

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DEPARTMENT OF COMPUTER APPLICATION

ACADEMIC SESSION: MARCH 2026 TO JUNE 2026

IV SEMESTER MCA

PROJECT SYNOPSIS

Title of the Project: “script to video generator using python and AI tools”

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1. Abstract :

The increasing demand for multimedia content in education, media, and information dissemination has created a need for efficient and accessible video production tools. This research presents VisoGen, an AI-driven system designed to automate the video content creation process.

A system based on artificial intelligence produces informative videos from one input: a topic name. Upon receiving the topic, the system uses Natural Language Processing (NLP) models to automatically retrieve and summarize key information. This generated text is then converted into speech using Text-to-Speech (TTS) technology.

Simultaneously, a visual representation of the text is created in the form of subtitles, which are synchronized with the audio for enhanced accessibility. The system then compiles the audio and visual components into a coherent video output. By automating research, content generation, voiceover, and video editing, the proposed system streamlines content creation, making it an efficient and valuable tool for educational, informational, and media production applications.

The system significantly reduces the time and effort typically required for manual video creation, offering a seamless solution for users to generate high-quality, multimodal content tailored to their needs. This approach can be particularly beneficial in various sectors such as online learning, media, marketing, and content creation, enhancing both productivity and accessibility.

2. Introduction

In the modern era, video content has become the most dominant form of communication due to its ability to convey information effectively and engagingly. Platforms such as YouTube, Instagram, and online learning systems rely heavily on video-based content. However, the process of creating videos manually is complex and involves multiple stages.

The Script to Video Generator is an AI-powered system designed to simplify this process. It enables users to convert textual scripts into videos without requiring advanced technical skills. The system uses intelligent algorithms to understand the content and generate visuals, audio, and subtitles automatically.

3. Problem Statement

Traditional video creation faces several limitations:

- High production cost
- Time-consuming process
- Requirement of skilled professionals
- Lack of scalability for bulk content creation

There is a need for an automated solution that can:

- Convert scripts into videos
- Generate voice narration automatically
- Select appropriate visuals
- Synchronize all multimedia components

4.Objectives

The objectives of this project are:

- To develop an automated system for script-to-video conversion
- To use NLP techniques for text understanding
- To generate realistic voice narration using TTS
- To map script content with relevant visuals
- To create subtitles synchronized with audio
- To provide a simple and user-friendly interface

5.Scope of the Project

The project has wide applications:

Education

- Convert lecture notes into video lessons

Marketing

- Create promotional videos

Social Media

- Generate reels and short videos quickly

News Industry

- Convert articles into video summaries

E-learning Platforms

- Automate content creation

6.Literature Survey

Various research works contribute to this domain:

- NLP techniques help in understanding textual content
- Deep learning models improve accuracy in content mapping
- GAN-based models are used for video generation
- TTS systems provide natural voice output

However, existing systems often lack full automation and integration of all components.

7.Proposed System

The proposed system is a fully automated pipeline consisting of multiple modules.
Modules:

- **Input Module**

Accepts text script

- **Text Processing Module**

Uses NLP to analyze content

- **Scene Generator**

Divides script into scenes

- **Media Retrieval Module**

Fetches images/videos

- **Text-to-Speech Module**

Generates voice narration

- **Subtitle Generator**

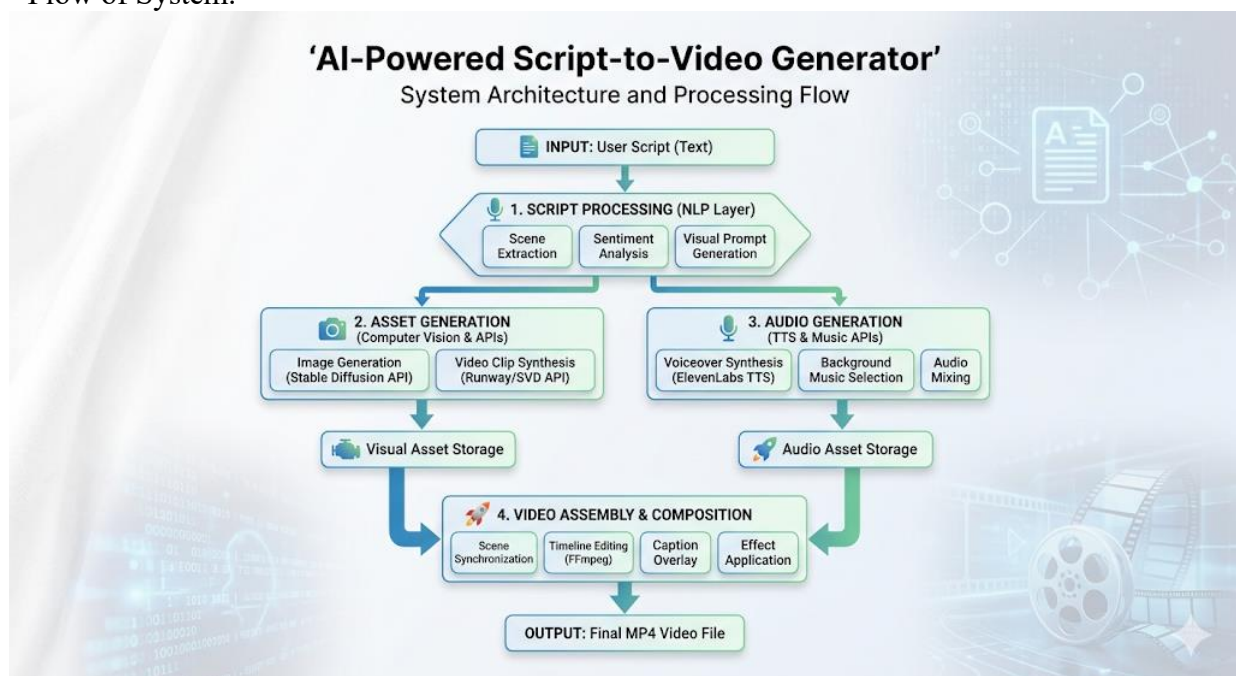
Creates captions

- **Video Assembly Module**

Combines all elements

8. System Architecture

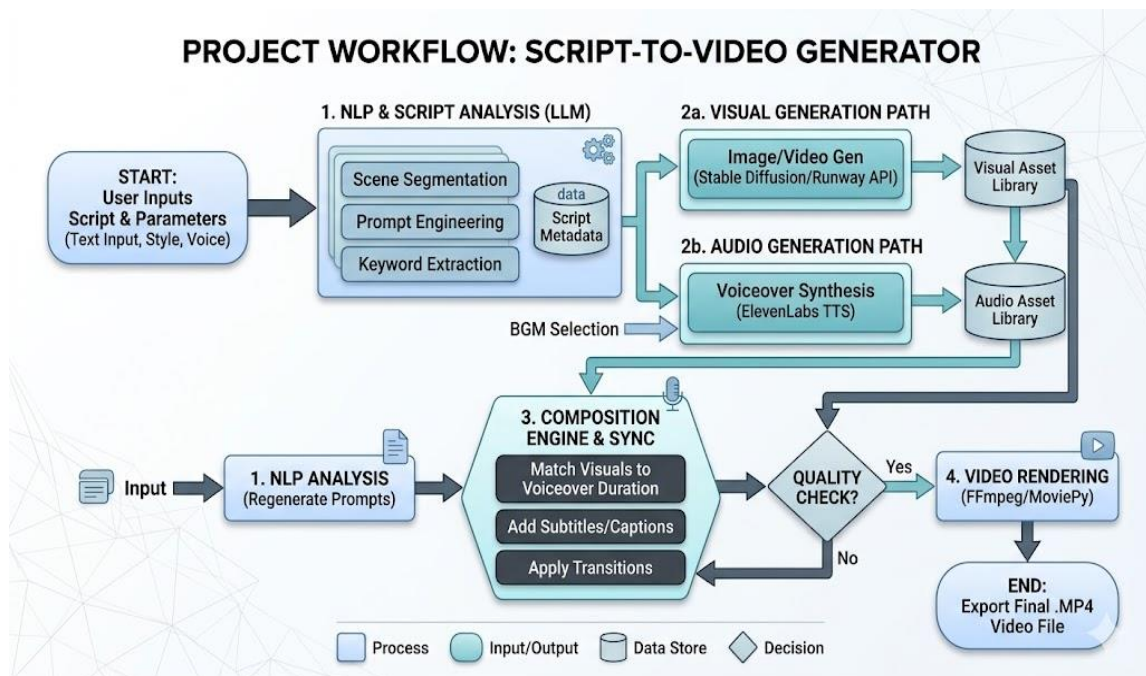
Flow of System:



9. Methodology

The system works in the following steps:

1. User inputs script
2. NLP processes text
3. Keywords are extracted
4. Script divided into scenes
5. Media is retrieved using APIs
6. Voice is generated using TTS
7. Subtitles are created
8. Final video is generated



Methodology: AI-Driven Script-to-Video Synthesis

The development of the Script-to-Video Generator follows a structured pipeline that integrates natural language understanding, generative AI, and automated digital media production. The methodology is divided into the following phases:

Phase 1: Input Processing & Narrative Analysis

The process begins with the ingestion of user-provided script text and creative parameters (e.g., tone, style, character preferences).

- **NLP Pipeline:** The system utilizes an Large Language Model (LLM) to perform **Tokenization** and **POS Tagging**, breaking the script into logical "scene chunks."
- **Storyboarding & Prompt Engineering:** The LLM automatically generates context-aware visual prompts for each scene. This ensures that the generated visuals align with the narrative intent (e.g., determining lighting, mood, and object placement).

10. Theory

Natural Language Processing (NLP)

NLP is a field of AI that gives computers the ability to read, understand, and derive meaning from human language. In a video generator, it acts as the "interpreter" of your script.

- **Tokenization:** The process of breaking a stream of text into smaller units called tokens (usually words or sentences). This allows the computer to process the script piece by piece.
- **POS (Part-of-Speech) Tagging:** Identifying the grammatical role of each word (e.g., noun, verb, adjective). This helps the AI understand the structure and intent of a sentence (e.g., distinguishing between "run" as a verb versus a noun).
- **Named Entity Recognition (NER):** A process that detects and classifies key information in text, such as names of people, locations, organizations, or dates. This is vital for choosing the correct imagery for a scene.

Machine Learning (ML) & Deep Learning (DL)

- **Machine Learning:** A branch of AI where algorithms learn patterns from data to make predictions or decisions without being explicitly programmed for every scenario.
- **Deep Learning:** A sophisticated subset of ML using neural networks with many layers (inspired by the human brain). In video generation, DL models are used to "dream up" visual content based on text prompts.
- **Text-to-Speech (TTS):** A technology that converts written text into synthesized, natural-sounding human speech. It is used to turn your script into the voiceover narration of your video.

Computer Vision

Computer Vision is the field of AI that enables machines to "see" and interpret visual data (images and videos).

- **Role in Video Processing:** It helps the system understand the contents of frames, stabilize footage, recognize objects within a scene, and even generate or transform images based on your script.

Video Processing

This is the technical stage of assembling and editing the generated assets.

- **OpenCV:** A comprehensive library for real-time computer vision. It is used to manipulate video frames, detect features, and perform image processing tasks.
- **MoviePy:** A Python library designed for video editing. It is used to cut clips, concatenate them, add text overlays, and adjust video timing.
- **FFmpeg:** A powerful, low-level multimedia framework. It acts as the "workhorse" for decoding, encoding, transcoding, and streaming audio and video files.

2. Tools and Technologies Definitions

Tool/Library	Definition & Purpose
Python	The primary programming language used for AI development due to its vast library support and readable syntax.
NLTK / spaCy	Powerful NLP libraries. NLTK is great for education and research; spaCy is optimized for production-level speed and industrial applications.
TensorFlow / PyTorch	Major deep learning frameworks used to build, train, and deploy the AI models that generate images and video.
gTTS	Google Text-to-Speech; a library that interfaces with Google Translate's speech-to-text API to generate audio from text.
APIs (Pexels/Unsplash)	These are stock media platforms. In a generator, the system uses their APIs to programmatically fetch high-quality, royalty-free images or video clips to match your script.

11. Advantages

- Saves time and cost
- Easy to use
- Fully automated
- Scalable

12. Limitations

- Limited creativity
- Depends on dataset quality
- Internet dependency

13. Expected Outcomes

- Automated video creation
- High-quality narration
- Accurate subtitles
- Faster production

14. Future Enhancements

- AI avatars
- Multilingual support
- Real-time video generation
- Emotion-based voice

15. Conclusion

The Script to Video Generator is a powerful AI-based system that simplifies video creation by automating the conversion of text into engaging multimedia content. It reduces manual effort, improves efficiency, and enables scalable content creation. With future advancements, it can revolutionize the content creation industry.

16. References

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